

GR 2 Lecture Note

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Abstract

This is the lecture note of the GR 2 course in EPFL 2026 spring semester. Mainly covering the topics in cosmology

Contents

1 Lecture 1: FRW Metric Spactime	1
1.1 Space Slices Metric	1
1.1.1 S^3 3-sphere	2
2 Lecture 2: Friedmann Equations and Solutions	3
Bibliography	4

1 Lecture 1: FRW Metric Spactime

For a cosmological model, we need to make some assumptions about the universe. The most common assumption is the cosmological principle:

Theorem 1.1

Cosmological Principle

The universe is homogeneous and isotropic on large scales.

The cosmological principle strongly constains the form of the metric. We assume that we can select a space slice at each time in some coordinate system, and consider the induced metric on each time slice.

1.1 Space Slices Metric

The cosmological principle requires that the metric on each time slice is homogeneous and isotropic, which means that the metric on each time slice is a maximally symmetric space. A maximally symmetric space is a space that has the maximum number of symmetries, which means that the Riemann curvature tensor has the following form:

$$\begin{aligned} R^{(3)} &= \text{constant} \\ R_{ij} &= \frac{1}{3} R^{(3)} g_{ij} \\ R_{ijkl} &= \frac{1}{6} R^{(3)} (g_{ik} g_{jl} - g_{il} g_{jk}) \end{aligned} \tag{1}$$

Thus we can see that a spactial metric can be classified into three types, and in fact each type of metric can be written in a simple form:

- $R^{(3)} = 0$, the metric is $\mathbb{R}^3$ 3-plane
- $R^{(3)} > 0$, the metric is S^3 3-sphere
- $R^{(3)} < 0$, the metric is $\mathbb{H}^3$ 3-hyperbolic space (Euclidean AdS space)

The metric of $\mathbb{R}^3$ is trivial, we will consider the other two cases in the following:

1.1.1 $\mathbb{S}^3$ 3-sphere

The metric of a 3-sphere can be given by inducing from a 4-dimensional Euclidean space with Cartesian coordinates x^i and the metric $ds^2 = \delta_{ij} dx^i dx^j$. We can induce a coordinate and metric by the following constraint and parametrization:

- Constraint:

2 Lecture 2: Friedmann Equations and Solutions

Bibliography